

LLM Agent for Interactive Data Analytics

Final Presentation

CS6604 – Adv Topics

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Research Question: Use cases of LLM Agent

1. Chat Assistant

- HITL process
- Tagging
- Tool selection

2. Reasoning Engine

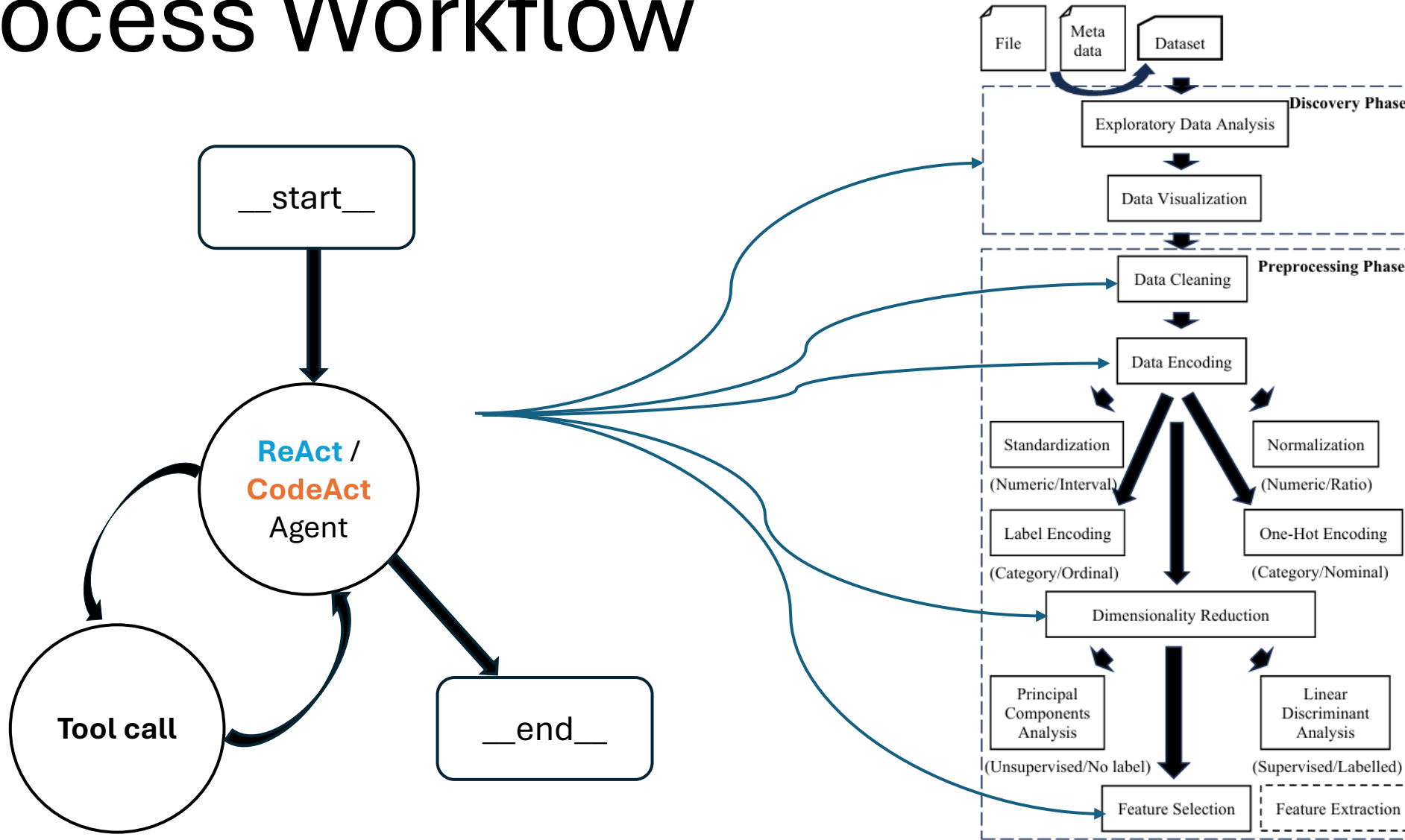
- Variable detection
- Feature info
- Filter Thresholds
- Decision-making
- Summarization(*)
- Visual analysis(*)

3. Code generation

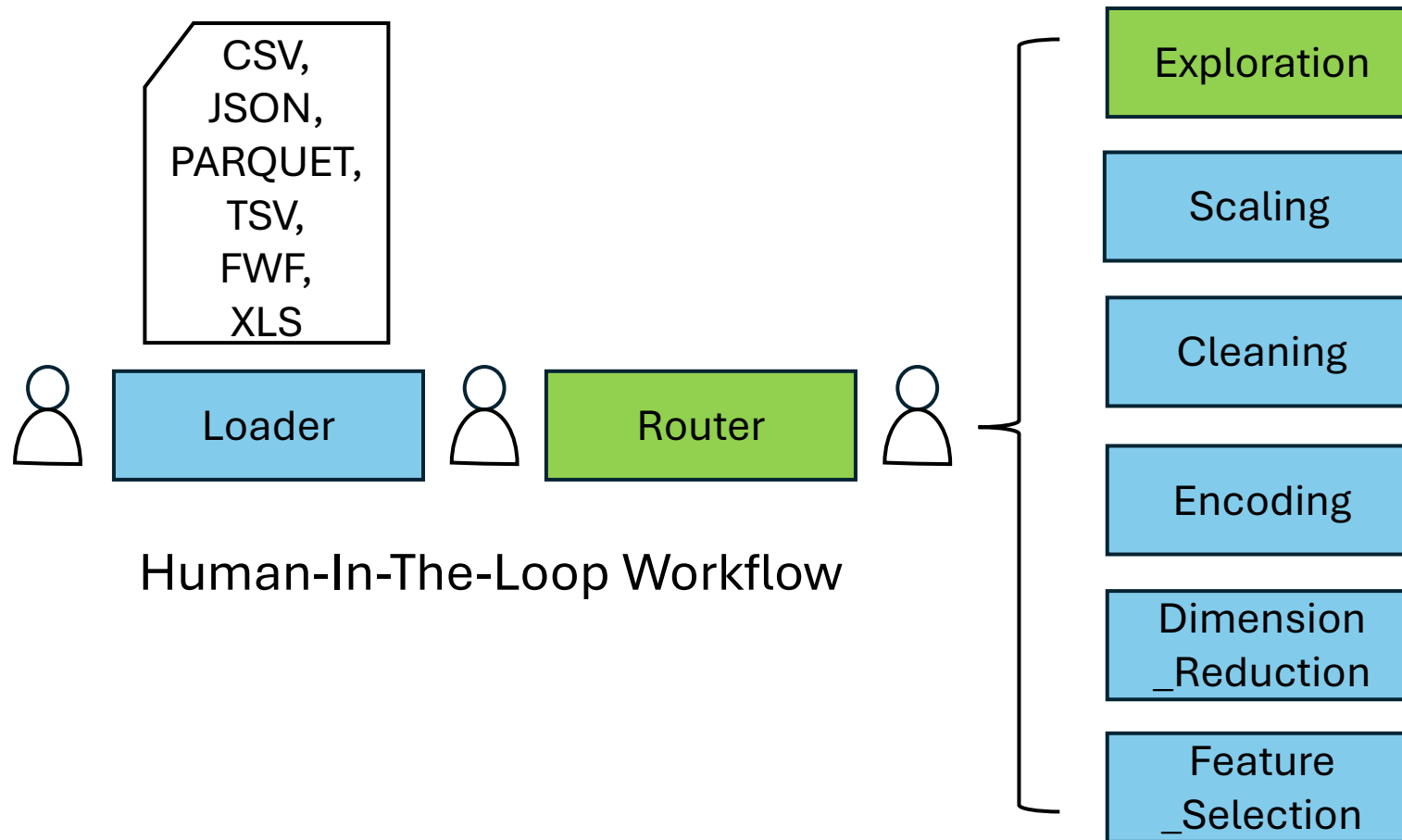
- Data Exploration
- Manipulation(*)
- Feature Extraction(*)

- The features marked (*) were initially proposed but were finally not implemented in the version due to restriction of scope of project

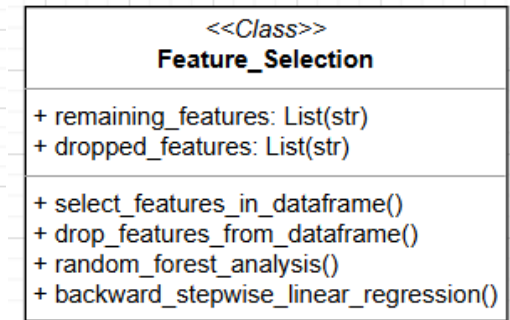
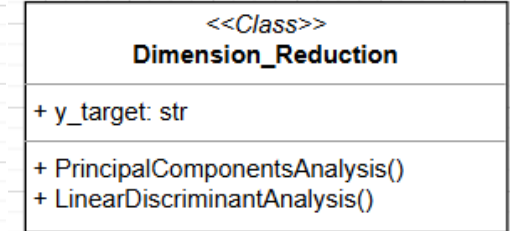
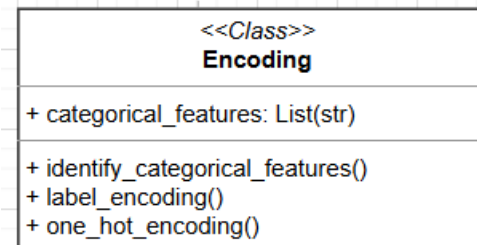
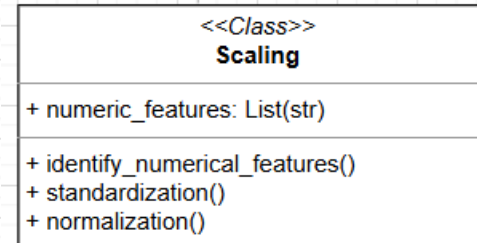
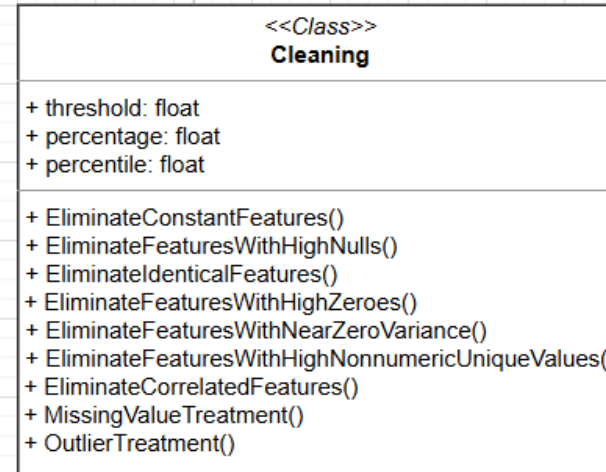
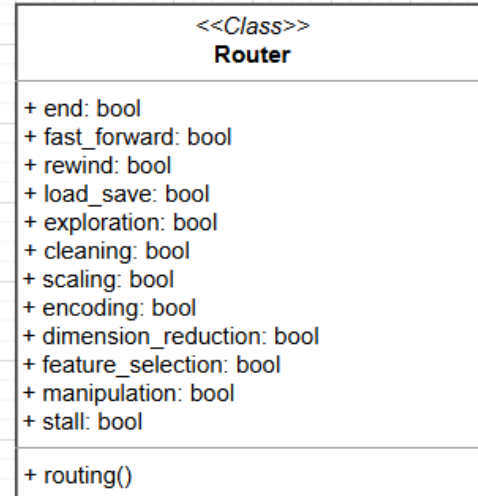
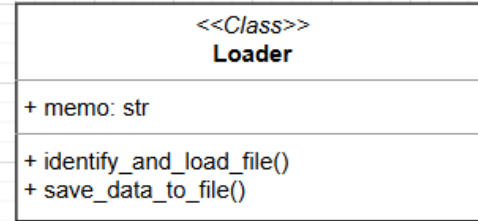
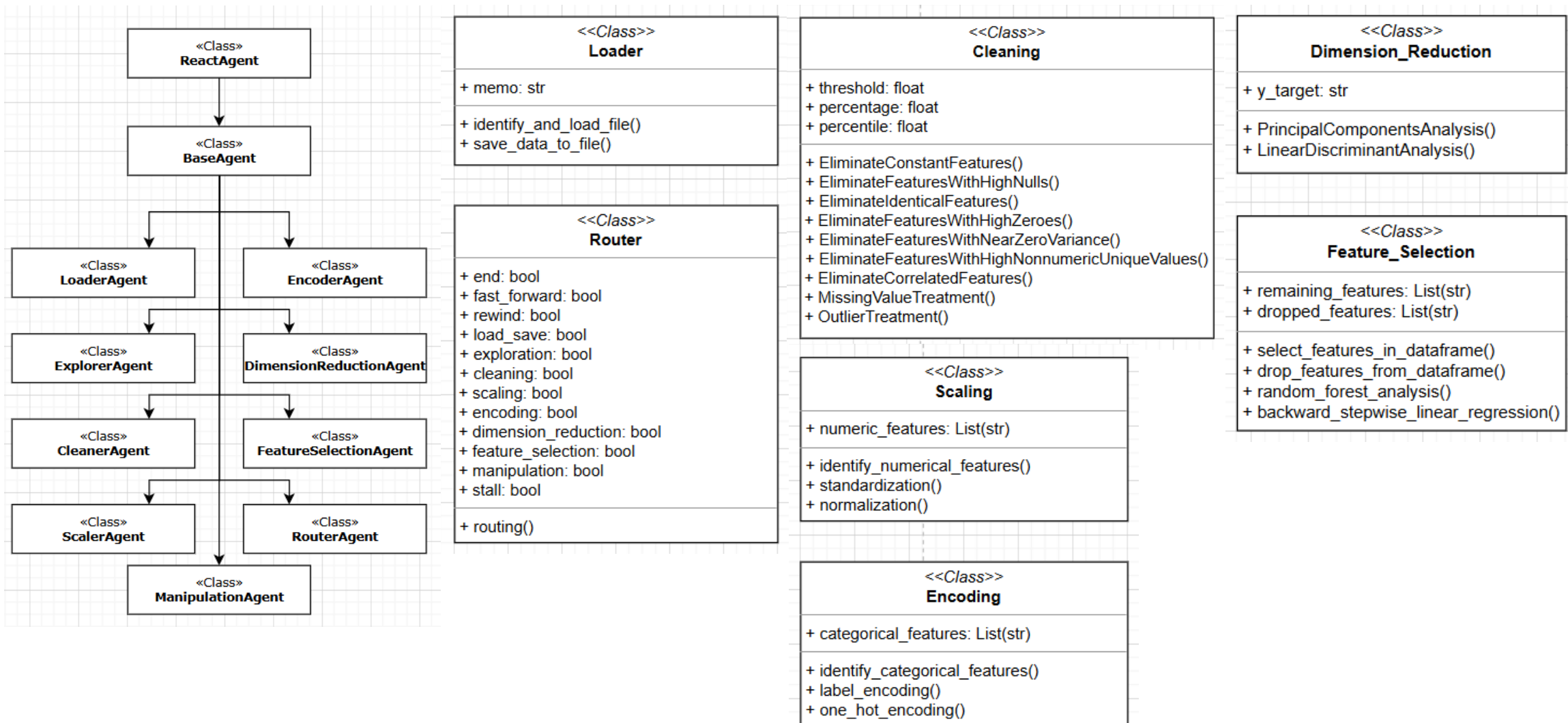
Process Workflow



Agentic Workflow & Roles – ReAct



Implementation: UML Class Diagrams of Toolset



Results

- For Encoding, there is a significant contribution in providing context assistance via ‘memo’ containing data descriptions and class information for categories to ascertain ordinality
- In all the other cases, the reasoning is working as intended without memorization affection the results.

	Manual Interventions (by user)	Scaling (Standardization /Normalization)	Encoding (Label/ One-hot Encoding)	Feature Selection (Random Forest)	Dimension Reduction (PCA)
Original dataset with ‘memo’	17/28 (60.7%)	22 / 28 (78.5%)	20/25 (80.0%)	21/23 (91.3%)	22/23 (95.6%)
Hidden dataset w/o ‘memo’	18/28 (64.3%)	21 / 28 (75.0%)	13/25 (52.0%)	20/23 (87.0%)	22/23 (95.6%)

Template and Tool Design Challenges

- Scaling/Encoding

- Multiple data frames:
 - test, train, validation together
- Inverse transform at a later stage
 - Save the y-encoder
- Time series (numerical)
 - multi column scaling

- Cleaning

- Null value treatment
 - Manual exception
- Feature elimination
 - Manual exceptions

Thank You